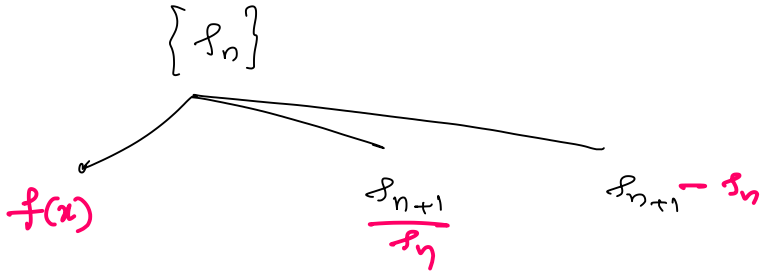
Lecture 22

Monday, 28 September 2026 8:14 AM

Recap: Cgs of $\{s_n\}$:

- If $\{s_n\}$ is monotonically ↑ (1) & bdd above then $\{s_n\}$ Cgs (2)



Today

Infinite Series:

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$$

Convergence of Series:

$$s_1 = a_1$$
$$s_2 = a_1 + a_2$$
$$s_3 = a_1 + a_2 + a_3$$
$$\vdots$$
$$s_n = a_1 + a_2 + a_3 + \cdots + a_n$$

$\{s_n\}$ → sequence of partial sums

Def: $$\sum_{n=1}^{\infty} a_n = \lim_{n \to \infty} s_n$$

Eg1: $$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots$$

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

$$a_n = \frac{1}{n(n+1)}$$

$$S_1 = a_1 = \frac{1}{1\cdot 2}$$

$$S_2 = a_1 + a_2 = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3}$$

$$S_3 = a_1 + a_2 + a_3 = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4}$$

$$\vdots$$

$$S_n = a_1 + a_2 + \cdots + a_n = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{n\cdot(n+1)}$$

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} S_n$$

$$S_n = \frac{2-1}{1\cdot 2} + \frac{3-2}{2\cdot 3} + \cdots + \frac{(n+1)-n}{n(n+1)}$$

$$= 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{n} - \frac{1}{n+1}$$

$$S_n = 1 - \frac{1}{n+1}$$

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{n\to\infty} S_n$$

$$= \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right)$$

$$= 1$$

Telescoping series:

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (b_n - b_{n+1}) = \ ?$$

$$S_1 = a_1 = b_1 - b_2$$

$$S_2 = a_1 + a_2 = (b_1 - b_2) + (b_2 - b_3) = b_1 - b_3$$

$$\vdots$$

$$S = a_1 + a_2 + \cdots + a_n = b_1 - b_{n+1}$$

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (b_n - b_{n+1}) = \lim_{n\to\infty} s_n$$

$$= \lim_{n\to\infty} (b_1 - b_{n+1})$$

$$= b_1 - \lim_{n\to\infty} b_{n+1}$$

Note: The telescopy series Cgs if $\lim_{n\to\infty} b_{n+1}$ is finite

$1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4}$

Be careful!

Cgs of $\sum_{n=0}^{\infty} x^n = ?$

"x is fixed number"

$s_1 = x^0 = 1$

$s_2 = x^0 + x^1 = 1 + x$

$s_3 = 1 + x + x^2$

$\vdots$

$s_n = 1 + x + x^2 + \cdots + x^{n-1}$

$$\sum_{n=0}^{\infty} x^n = \lim_{n\to\infty} s_n$$

$$\sum_{n=0}^{\infty} x^n = \lim_{n\to\infty} s_n = \lim_{n\to\infty} \frac{1-x^n}{1-x}$$

$s_n = 1 + x + x^2 + \cdots + x^{n-1}$

$x s_n = x + x^2 + \cdots + x^n$

$\Rightarrow s_n - x s_n = 1 - x^n$

$\Rightarrow s_n = \frac{1-x^n}{1-x}$ $\quad x \neq 1$

If $-1 < x < 1$,

$$\lim_{n\to\infty} \frac{1-x^n}{1-x} = \frac{1}{1-x} - \lim_{n\to\infty} \frac{x^n}{1-x}$$

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad \text{if} \quad -1 < x < 1$$

$\sum_{n=0}^{\infty}$

If $x=1$ then $\sum_{n=0}^{\infty} x^n$ dgs

$= \lim_{n\to\infty} s_n = \lim_{n\to\infty} (n)$

↓ not finite

If $x=-1$ then $\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} (-1)^n$

$s_1 = 1$

$s_2 = 1-1 = 0$

$s_3 = 1, \quad s_4 = 0, \ldots$

$$s_n = \begin{cases} 1 & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

$\lim_{n\to\infty} s_n$ doesn't exist

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \qquad -1 < x < 1$$

Geometric series

Eg1: Discuss the cgs of $\sum_{n=1}^{\infty} \frac{2^n+3^n}{6^n} = \frac{1}{2}+1$:

$\sum_{n=1}^{\infty} \frac{2^n}{6^n}$, $\sum_{n=1}^{\infty} \frac{3^n}{6^n}$

↓

$\sum_{n=1}^{\infty} \frac{2^n}{6^n} + \frac{2^0}{3^0} - \frac{2^0}{3^0}$

$\left(\sum_{n=1}^{\infty} \frac{2^n}{6^n}\right) \qquad \frac{2^0}{3^0} = \sum_{n=1}^{\infty} \left(\frac{2}{6}\right)^n - 1$

Thm: If $\sum_{n=1}^{\infty} u_n$ & $\sum_{n=1}^{\infty} v_n$ cgs then $\sum_{n=1}^{\infty} (\alpha u_n + \beta v_n)$ cgs

Check

1

$$= \left(\sum_{n=0} \frac{1}{6^n} \right) - 3^0$$

$$= \frac{1}{1 - \frac{2}{6}} - 1$$

$$= \frac{1}{2}$$

Necessary Condition for Cgs:

If $\sum_{n=1}^{\infty} a_n$ cgs then

(OR)

If $a_n \to 0$ as $n \to \infty$ then $\sum a_n$ dgs

P: $\sum_{n=1}^{\infty} a_n$ cgs

Q: $a_n \to 0$ as $n \to \infty$

$P \Rightarrow Q$

$\Leftrightarrow$

$\sim Q \Rightarrow \sim P$